PROJECT PROPOSAL

PEERGRID: A College Discovery and KEAM Rank-Based Admission Guidance Social Platform

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***Abstract*— Choosing the right undergraduate engineering or professional college after the Kerala Engineering Architecture Medicine (KEAM) entrance examination is a decision that most students and parents in Kerala make with incomplete and scattered information. Existing sources such as college websites, coaching-centre pamphlets, and informal social-media groups rarely present a unified, comparable, and rank-aware picture of colleges. This proposal presents PeerGrid, a college-connectivity mobile and web application designed exclusively for the Kerala higher-education ecosystem. PeerGrid combines (i) a structured, parameter-based directory of colleges covering academic performance, placement records, accreditation, infrastructure, and extracurricular activity, with (ii) a machine-learning-based recommendation engine that suggests suitable colleges to a student based on their KEAM rank, category, preferred branch, and historical allotment trends, and (iii) a closed, verified social layer that lets admitted and prospective students connect, ask questions, and build peer communities around specific colleges and departments. This document outlines the problem being addressed, the proposed objectives, the technical innovation, the technology stack, an implementation plan with a project timeline, and the future scope of the platform.**

I. INTRODUCTION

Every year, more than a hundred thousand students in Kerala appear for the KEAM entrance examination for admission to engineering, pharmacy, and related professional degree programmes. Once a student's rank is announced, they must make a high-stakes decision within a short allotment window: which colleges to list, in what order, and for which branch. This decision materially affects the quality of education, placement opportunities, and campus experience the student will receive for the next four years.

At present, students rely on a mixture of outdated government brochures, unofficial WhatsApp and Telegram groups, coaching-centre advice, and word of mouth from seniors to make this decision. There is no single, trustworthy, and personalised platform that brings together verified college data, historical rank-cutoff trends, and a peer community in one place. PeerGrid is proposed to fill this gap. It is a college-connectivity social application built specifically for the Kerala student community, which (a) presents a ranked, filterable directory of colleges with transparent parameters, (b) uses a machine-learning recommendation engine to suggest colleges suited to a student's KEAM rank and preferences, and (c) provides a moderated social space where students, seniors, and alumni of a given college can interact.

II. PROBLEM STATEMENT

The current college-selection process for KEAM-qualified students in Kerala suffers from the following gaps:

- Fragmented information: College data on academics, placements, faculty, and infrastructure is spread across many websites, PDFs, and social-media posts, with no common format for comparison.
- No rank-aware guidance: Students have limited access to historical allotment/cutoff-rank data, making it difficult to judge realistic chances of admission to a given college and branch.
- Lack of personalisation: Generic "top college" lists ignore an individual student's rank, category, location preference, and branch interest.
- Weak peer support: Newly admitted or prospective students have no verified way to connect with current students or alumni of a specific college before making a decision.
- Trust and verification issues: Informal groups and forwarded messages often carry inaccurate or outdated information, leading to poor decisions.

PeerGrid addresses these problems by combining structured, parameter-driven college data with a machine-learning recommendation engine and a verified social layer within a single application.

III. OBJECTIVES

The primary objectives of the PeerGrid project are:

- To design and maintain a structured database of Kerala colleges with measurable parameters such as academic performance, NAAC/NBA accreditation, placement statistics, faculty strength, infrastructure, and extracurricular/event activity.
- To develop a machine-learning-based recommendation system that suggests a ranked list of suitable colleges and branches to a student based on their KEAM rank, category, and stated preferences.
- To build a social networking layer restricted to verified college communities, enabling current students, alumni, and incoming students to interact, ask questions, and share experiences.
- To provide historical rank-cutoff and allotment-trend visualisations that help students set realistic expectations during option registration.
- To create a mobile-first, easy-to-use interface tailored to first-generation college applicants and their parents.
- To ensure data authenticity through a college/administrator verification workflow and community moderation.

IV. INNOVATION

While several general-purpose education portals exist in India, none combine rank-specific machine-learning recommendations with a closed social layer focused exclusively on the Kerala KEAM admission cycle. The key innovative aspects of PeerGrid are:

- Rank-aware recommendation engine: Unlike static "best college" lists, PeerGrid's model treats the student's KEAM rank, category, and branch preference as first-class inputs,

producing a personalised, probability-weighted shortlist of colleges.

- Hybrid recommendation approach: The system combines content-based filtering (matching college parameters to student preferences) with collaborative filtering (learning from the choices and outcomes of past students with similar ranks) to improve suggestion quality over time.
- Integrated social graph: College-specific communities are unlocked based on verified admission or alumni status, reducing spam and misinformation compared to open social-media groups.
- Kerala-specific focus: The platform is purpose-built around the KEAM examination structure, option-registration process, and the specific set of government, aided, and self-financing colleges in Kerala, rather than being a generic pan-India tool.

V. TECHNOLOGY STACK

PeerGrid is designed as a three-tier application consisting of a mobile/web client, an application server, and a machine-learning recommendation service. Table I summarises the proposed technology stack.

TABLE I PROPOSED TECHNOLOGY STACK

Layer	Technology
Mobile / Web Client	Flutter (Android & iOS), React.js (admin/web dashboard)
Backend / API	Node.js with Express.js, RESTful APIs, JWT-based authentication
Database	PostgreSQL (structured college & user data), MongoDB (chat/social feed data)
Recommendation Engine	Python, scikit-learn, Pandas, NumPy (content-based + collaborative filtering models)
Model Serving	FastAPI micro-service exposing the trained model to the main backend
Cloud & DevOps	AWS (EC2, S3, RDS) or Firebase, Docker for containerisation, GitHub Actions for CI/CD
Real-time Features	Firebase Cloud Messaging / Socket.IO for notifications and chat
Data Sources	KEAM/CEE official allotment data, college-provided data, NAAC/NBA public reports

A. Recommendation Model Design

The recommendation engine is planned around a two-stage pipeline. In the first stage, a content-based filtering model scores every college against a student's KEAM rank, category, home district, and branch preference using normalised college parameters (placement percentage, faculty–student ratio, accreditation grade, and event/activity index). In the second stage, a collaborative filtering component, trained on anonymised historical option-registration and allotment outcomes of previous batches, refines the shortlist by learning patterns such as "students with a similar rank and category who chose branch X ultimately preferred college Y." The two scores are combined into a final ranked list presented to the student, along with an estimated admission likelihood for each college–branch combination.

VI. IMPLEMENTATION PLAN AND TIMELINE

The project is planned to be executed in eight phases over a period of approximately sixteen weeks, as summarised in Table II.

TABLE II IMPLEMENTATION PLAN AND TIMELINE

Phase	Activity	Week
1	Requirement gathering, literature survey, and study of KEAM allotment process	1–2
2	System design: database schema, API design, and UI/UX wireframing	2–3
3	College-data collection and verification (academics, placements, accreditation, events)	3–5
4	Backend development: authentication, college directory API, social-feed API	4–7
5	Machine-learning model development: data preprocessing, content-based and collaborative filtering models, evaluation	5–9
6	Frontend/mobile app development and integration with backend and ML micro-service	6–11
7	Integration testing, user-acceptance testing, and bug fixing	11–13
8	Beta deployment, user feedback collection, and iterative refinement	13–16

This phased approach allows the college-data collection and backend work to proceed in parallel with early machine-learning experimentation, so that a working prototype of the recommendation engine is available for integration testing by week nine, leaving adequate time for refinement before beta launch.

VII. FUTURE SCOPE

Beyond the initial release, PeerGrid can be extended in several directions:

- Expansion to other state-level entrance examinations (e.g., other state CETs) beyond KEAM, and eventually to national examinations such as JEE and NEET counselling.

- An AI-based chatbot counsellor to answer frequently asked questions about colleges, courses, and the option-registration process in real time.
- A verified alumni-mentorship module connecting current students with alumni working in relevant industries for career guidance.
- Predictive analytics on year-on-year cutoff-rank trends to help future applicants plan option lists more accurately.
- A scholarship- and hostel-recommendation module integrated with the core college recommendation engine.
- Partnerships with colleges for officially verified data feeds, reducing reliance on manually curated information.

VIII. CONCLUSION

PeerGrid proposes a focused, technically grounded solution to a real and recurring problem faced by hundreds of thousands of KEAM-qualified students in Kerala every year: making an informed, personalised, and confident college choice. By combining a structured college-parameter database, a machine-learning-based recommendation engine, and a verified peer-to-peer social layer, the platform aims to reduce information asymmetry, improve decision quality, and build a supportive student community around the college-admission journey. The phased implementation plan outlined in this proposal provides a realistic path from requirement gathering to a testable beta release within sixteen weeks.

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